# Shapir Laboratory Test Results

eco BRICK lightweight thermal concrete for use beneath flooring as an underfloor infrastructure layer

| Test mixture | Test date | eco BRICK aggregate | Fresh density | Air content |
|---|---|---|---|---|
| High-strength eco BRICK mixture | 07 March 2023 | 350 kg/m³ | 1,811 kg/m³ | 7% |

| 660 | 9.115 | 14.535 | SF-2 |
|---|---|---|---|
| flow / spread | MPa - 3 day average | MPa - 7 day average | flow class |

## Mixture composition

| Component | Supplier / source | kg/m³ |
|---|---|---|
| eco BRICK aggregate | Etziona | 350 |
| Shapir sand 0-4 | Etziona | 400 |
| Sea sand | Rishon LeZion | 450 |
| Fly ash | - | 100 |
| Cement | Vasiliko | 420 |
| Water | - | 220 |
| S-500 admixture | - | 7.02 |

## Laboratory observations and strength results

| Parameter | Recorded result |
|---|---|
| Fresh density | 1,811.43 kg/m³ |
| Air content | 7% |
| Flow / spread | 660 (SF-2) |
| 3-day compressive strength | 9.115 MPa average |
| 7-day compressive strength | 14.535 MPa average |
| 28-day compressive strength | Not recorded in supplied workbook |
| Workability note | Not sufficiently self-leveling |

*Source note: Prepared from the supplied Excel workbook "Shapir-Laboratory-Test-Results.xlsx". Values are reproduced as recorded. No value has been inferred where the workbook contains blank or unresolved formula fields.*